# BuyerUnion | Complete Project PRD v4

CalHacks 13.0 | Revised October 23, 2026 | Product, engineering, economics and sponsor reference | No length limit

## 1. Core Details and Why

**Build:** a marketplace for manufactured consumer products: buyers write a natural-language prompt; a neutral assistant confirms its meaning; merchant agents find stock, make a fit argument and compete through authorized variable prices; a bounded council challenges claims; independent web evidence informs the final decision. Technology, household accessories, appliances and clothing share this flow. Laptops are a demonstration, not the product boundary. Nothing is bought or fulfilled in the real world during the demo.

**Confirmed scope:** four teammates, two own UI in Figma; no existing application; manufactured products, new/refurbished where applicable; U.S./USD; San Francisco delivery; buyer-selected deadline; disclosed mock merchant stock/terms; real web evidence; Visa sandbox attempted, labeled simulation otherwise. No language/framework preference. This supersedes both real-inventory-first and laptop-only assumptions.

**Chosen defaults:** TypeScript, React/Vite, Convex/Auth/Workflow, direct Gemini API, Apify; actual frontend hosted on Vultr. Three fictional merchants; one primary product type and up to two related product types per request, all from one merchant; at most five offer cards. Each line quantity is 1-10. Category profiles supply typed rules; adding categories must not require changing the council, checkout or screen sequence. No second database, real-money mode or multi-seller checkout.

**Coverage boundary:** the architecture accepts extensible product categories; launch profiles cover laptops/tech accessories, clothing, household accessories and small appliances. Do not claim universal verified inventory or evidence coverage. Unknown categories stay editable drafts until a validated profile exists. Services, consumables and regulated goods are outside this prototype; large-appliance installation is not promised.

**Why:** reduce buyer comparison work and merchant quotation effort while creating profitable opportunities to win a purchase. The merchant agent protects its seller's contribution; the buyer assistant protects the buyer's stated requirements and approval. The platform cannot promise to maximize both sides' surplus simultaneously. Debate is useful only when it improves a valid offer or resolves a material claim. Longer reasoning is not evidence of a better outcome.

| Visa criterion | Required demonstration |
|---|---|
| Innovation 25% | Tailored merchant offers and a challenge that produces a useful correction/concession; avoid claiming to invent RFQs. |
| UX 25% | Broad and precise input, editable requirements, concise progress, understandable offers and clear purchase approval. |
| Technical 20% | Actual scoped tool calls, enforced policies, real source retrieval, persisted state and recovery. |
| Commerce 15% | Measure comparison/quotation effort and fulfilled demo offers; no invented savings or conversion uplift. |
| Trust 15% | Mock disclosure, separate commercial/private data, applicable citations, truthful incomplete states and bounded approval. |

**Readiness boundary:** this document resolves product and architecture choices sufficient to begin the demo. It does not certify provider access, scraping success, measured performance, or prize eligibility. External access and source/API smoke tests remain explicit gates. Live merchant adoption and economics remain future

validation questions, not prototype blockers.

**v4 changes:** removes the former ten-page limit; consolidates the project context and sponsor/prize strategy; makes prompt-first intake, commercial bidding, optional shared targets, neutral fit scoring, conditional final bids and independent sentiment explicit. Earlier reports remain research background; this document governs the current implementation scope. **Reading guide:** overview/decisions and buyer flow first; commercial mechanics in sections 7-10; implementation in 11-15 and 23-25; sponsor strategy in 16-19; research/risks/metrics in 20-22; external gates and sources in 26-27. Section numbers are not PDF page references.

## 2. Problem, user sentiment and product rationale

**Buyer problem:** shoppers repeatedly translate an everyday need into product filters, compare variants and delivered prices, judge reviews, and check seller terms. The product should let people describe the outcome once and compare a few defensible offers. **Merchant problem:** serving a relevant buyer can require catalog search, stock checks, discretionary pricing and explanation; the agent acts as a controlled automated sales desk.

Earlier research examined selected public shopper discussions and primary research. It supports comparison friction and trust concerns, not proven demand for this marketplace. No representative sentiment scrape, merchant commitment, original interview study or conversion experiment has been completed.

| Observed theme | Evidence and design implication |
|---|---|
| Verifying AI advice creates work | Gartner's May 2026 release reports that 54% of recent shopping-AI users double-checked all information. The underlying Nov-Dec 2025 survey had 846 U.S. consumers; this statistic concerns its AI-user subgroup. Provide inspectable evidence, not another unsupported recommendation. [R1] |
| Assistance differs from purchase authority | A separate January 2026 Gartner survey of 322 U.S. consumers found willingness to delegate purchase decisions reached only 11% in the more receptive categories studied. Keep final approval explicit. [R1] |
| Delivered cost and checkout friction matter | Baymard identifies extra charges and checkout usability as abandonment drivers. Show a comparable complete demo total before approval; do not translate a general research finding into a promised conversion uplift. [R2] |
| Durability research is fragmented | Selected BuyItForLife discussions describe cross-checking reviews, warranties and forums. Use exact-product evidence and preserve uncertainty. This is anecdotal evidence, not a prevalence estimate. [R3] |
| Clothing size labels are unreliable guides to fit | A selected PlusSize discussion describes inconsistent charts. Preserve brand/size-system and measurement context; do not translate every M into the same dimensions. [R4] |
| Returns and availability affect trust | Earlier research found accounts of refund uncertainty and advertised stock mismatch, with conflicting experiences. Distinguish an offer, a stock hold and a completed order; mock inventory cannot prove real fulfillment reliability. |

**User groups:** broad-intent shoppers needing interpretation; precise shoppers with hard requirements; merchants seeking profitable qualified demand; merchant operators defining authority. Their interests overlap but differ: a merchant agent represents its seller, and the buyer assistant should explain that clearly.

**Hypothesis:** a prompt-led competitive offer process can reduce research/quotation effort while improving acceptable purchase opportunities. Discounting alone is not the product; a reliable suitable offer and an informed decision are. The central market risk is whether enough relevant merchants participate for each category, location and deadline.

**Success story to prove:** a buyer writes an ordinary request; agents identify relevant variants, reject unsuitable stock, produce a bounded commercial improvement, substantiate the fit argument and present independent sentiment. The buyer can see why one offer is cheaper and another better meets a stated preference without being pushed into a purchase.

## 3. Decision register and scope boundaries

This register separates confirmed user directions from implementation defaults. Defaults are concrete choices for the build, not claims that the team has already implemented or tested them.

| Status | Decision |
|---|---|
| Confirmed | General manufactured-products marketplace; laptops only an example/demo. Include technology, clothing, household accessories and appliances in the model and fixtures. |
| Confirmed | Four people; two teammates handle UI in Figma. No initial application, merchant inventory or sample data. No framework/language constraint. |
| Confirmed | Custom buyer UI; skip Fetch AI if the primary workflow must run inside ASI:One. |
| Confirmed | Natural-language buyer prompt; merchant agents do catalog/product work, fit arguments, council and useful price competition; independent web sentiment closes the decision loop. |
| Confirmed | Disclosed mock inventory for the demo, because merchant authority is unlikely. Real evidence must be real, applicable and cited. |
| Confirmed | New/refurbished where applicable; U.S./USD, San Francisco demo area and buyer-selected deadline. Visa sandbox attempted, labeled simulation fallback. |
| Confirmed | Bidding is central; merchants may offer modest variable discounts to win profitable sales. One council round acceptable; more compute only when useful. Former PRD length cap removed. |
| Proposed default | Neutral extraction/confirmation precedes merchant interpretation; private maximum plus optional explicitly shared target. Merchant self-scores are advisory. |
| Proposed default | One initial private offer phase, one claims council, at most one sealed final-price revision phase; one extra verification-only council when justified. |
| Proposed default | TypeScript/React/Vite, Convex/Auth/Workflow, direct Gemini API, Apify and genuine Vultr frontend hosting. One primary database and orchestration path. |
| Proposed default | Three fictional merchants; maximum two active candidates each; one primary product type plus two related types per single-merchant bundle; line quantities 1-10. |
| Proposed default | Merchant-specific discount caps, starting at 5-10% where permitted; floor/cost guards override cap. Zero discount is allowed. |

**Prototype exclusions:** live-money purchases, real fulfillment promises, cross-merchant atomic checkout, escrow, automated returns/refunds, unsolicited seller outreach, user credit scoring, biometric fit prediction, voice input, uploaded-photo analysis, public agent registration and arbitrary external merchant agents. These are future extensions, not requirements hidden inside the first demo.

**Coverage is explicit:** launch category profiles prove the shared model; they do not prove every manufactured item has known specifications or usable reviews. Unsupported products can remain draft requests until a validated profile and supply exist. Services, consumables, regulated products and installation commitments require separate scope decisions.

**Terminology:** request = confirmed buyer need; candidate = merchant's proposed bundle; offer version = immutable proposed terms; council = claim/challenge/response; final bid = permitted commercial revision; fit assessment = neutral requirement comparison; sentiment = attributed external reports; firm quote = selected revalidated terms; reservation = stock allocation; approval = buyer authority; execution = one recorded payment/demo attempt.

# 4. Flow: screens, requirements and comparison

**Buyer screens:** prompt conversation -> plain-language confirmation -> progress -> comparison/evidence -> final quote -> result. The main input is a prompt, not a specification form. Preserve the request on reload. Show "Demo merchants and inventory; no real purchase" and evidence/payment mode badges. Merchant screens: own catalog, policy controls, pause toggle, own bids/rejections and permitted council events.

1. Establish an anonymous session and accept a prompt up to 4,000 characters: e.g., "I need a quiet kettle for a small apartment, under $60 delivered by Friday." A neutral intake agent extracts explicit needs and proposes interpretations. Merchant agents later map the same confirmed brief to their own catalogs; they do not privately redefine it.

2. Ask only material follow-ups in chat, at most three grouped turns. Show a short editable summary/chips: category, quantity, deadline, region, conditions, must-haves, preferences and price disclosure. Budget/deadline start unset; quantity one/profile condition defaults are visible. Do not require the buyer to know technical parameters. Optional advanced details are available. If ambiguity remains, save a draft and explain what needs an answer.

3. The buyer confirms the interpreted brief and private all-in maximum. An optional target is shared only with explicit permission. Persist a version; distribute a sanitized common brief, never the raw prompt containing private prices or personal context. Show stages, elapsed time and actual merchant participation. Edits create a new version; cancellation is allowed before execution.

4. Merchant agents inspect own inventory/policy, propose suitable products, permitted prices and evidence-backed arguments. Neutral checks produce comparable fit explanations. Run a bounded council, then one sealed final-price round only if useful. Research final 3-5 offers when available; show the true count otherwise. Cards separate fit, delivered price, terms and sentiment.

5. Buyer inspects sources and chooses. Revalidate/reserve only the selected bundle; show a three-minute server countdown, changed terms, and final approval. No inferred approval.

**Conversational rules:** "under $1,000" proposes a private ceiling; "around $1,000" needs clarification before becoming a hard maximum. "Try for $950" is a proposed shared target, not permission to disclose it until confirmed. A request for "quiet" or "good for college" remains a preference until grounded in supported facts. The buyer can correct the summary in natural language; no merchant argument can silently change it.

**No-match path:** show the limiting requirements and allow a new prompt or explicitly approved relaxation. Never invent sellers, consent, a stock commitment or a discount to keep the flow moving.

# 5. Category profiles and manufactured-product scope

Common constraints: category, brand/model if specified, quantities, condition, total ceiling, arrival, minimum seller warranty/returns. Product constraints are typed entries: line ID, profile field ID, allowed operator, value/unit, hard or soft. Pin the immutable profile version through validation/checkout; updates apply to new requests. Profiles define labels, types, ranges, units, operators, clarification, variant identity, compatibility and evidence scope. Models propose values; server validators enforce them.

| Profile | Launch-specific rules |
|---|---|
| Laptop / tech accessory | OS, RAM/storage GB, weight grams, USB-C charging; refurbished battery-health percent; explicit supported-device compatibility. |

| Profile | Launch-specific rules |
|---|---|
| Clothing | Garment type, brand-specific size label/system, color, material percentages, garment measurements in mm. No universal M-to-M equivalence; fit preference is advisory. Default new only. |
| Household accessory | Item type, dimensions mm, material, color, mounting/interface compatibility. Do not infer load capacity or attachment safety. Default new only. |
| Small appliance | Type, dimensions mm, capacity with named unit, voltage V/frequency Hz and plug/region. Compare measured noise/energy only with units and test basis. Default new/refurbished. |

Operators: equals/in/min/max/containsAll, allowed per field; convert compatible units before comparing. Reject unknown fields/operators or contradictory bounds. Unknown hard facts fail; subjective claims such as "flattering" remain preferences. New profiles are operator-reviewed configuration, never generated executable rules.

Store UTC milliseconds; show America/Los_Angeles. Date-only means 23:59:59 local, confirmed explicitly; reject past dates. Demo arrival = quote time plus configured calendar days. No street address/card data. UI must support keyboard use, readable status announcements and preserved focus.

**Extending coverage:** add an operator-reviewed profile, typed attributes, permitted conditions, variant keys, compatibility checks, source applicability rules and positive/negative fixtures. Expose the new profile through the existing prompt flow. Do not add a new negotiation or checkout engine. Generic descriptions can be drafted for any manufactured product; verified matching requires supported fields and actual inventory.

**Constraint provenance:** each extracted requirement records the source message span, whether explicit or proposed, and buyer confirmation. A merchant may infer useful candidates from the use case but must label assumptions, never present them as buyer requirements. Conditions such as refurbished battery health apply only to relevant categories.

# 6. Catalog, mock commerce and seed requirements

**Identity:** Product = family/model; Variant = exact configuration, size/color/material or regional version; Listing = merchant's stock/condition/terms for that variant. Manufacturer part number/GTIN is optional and never invented. Include category/profile version and per-attribute provenance: manufacturer, real report or synthetic demo. Stock belongs to listings, not families.

**Laptop pitch fixtures:** internal IDs, not manufacturer part numbers. Sources establish specifications, never stock. [S7-S10]

| ID | Canonical configuration | Source |
|---|---|---|
| P1 | MacBook Air M2 2022; 8-core CPU/8-core GPU; 16 GB/512 GB | Apple US M2 [S7] |
| P2 | MacBook Air 13-inch M3 2024; 8-core CPU/8-core GPU; 16 GB/512 GB | Apple US M3 [S8] |
| P3 | MacBook Air 13-inch M4 2025; 10-core CPU/8-core GPU; 16 GB/512 GB | Apple US M4 [S9] |
| P4 | Same M4 family; 10-core CPU/10-core GPU; 24 GB/512 GB | Separate configuration [S9] |
| P5 | ThinkPad T14 Gen 5 Intel; Core Ultra 5 125U; 16 GB/512 GB; Windows 11 Pro English | US model 21ML0053US; withdrawn status does not imply current stock [S10] |
| P6 | MacBook Air M2 2022; 8-core CPU/8-core GPU; 8 GB/256 GB | Low-spec fixture [S7] |

Retain source/checked date. Unknown attributes stay null. Apple OS family is macOS; no unsupported OS-version, weight or benchmark claims.

### Fixed commercial fixtures

Fictional USD dollars; store cents. N=new, R=refurbished. Entry = product / condition / quantity / list / floor.

| Merchant | Laptop listings | All-category demo policy |
|---|---|---|
| Demo Merchant A | P1/R/1/620/580; P3/N/2/990/950; P5/R/1/690/650; P6/R/1/480/450 | Shipping 15; arrival +3 calendar days; 12-month demo seller warranty; 14-day returns. |
| Demo Merchant B | P2/R/1/750/710; P3/N/1/1020/965; P5/N/1/850/810; P6/N/2/550/520 | Shipping 0; arrival +1 day; 24-month demo seller warranty; 30-day returns. |
| Demo Merchant C | P1/R/1/610/610; P4/N/1/1090/1090; P5/R/1/680/680; P6/R/1/460/460 | Shipping 0; arrival +2 days; 12-month demo seller warranty; 14-day returns. No price concession. |

Refurbished listings are unique units. Synthetic health: A/P1=92%, A/P5=90%, B/P2=88%, C/P5=85%; others unknown. These are not certified diagnostics. Unknown fails a mandatory health threshold.

Each merchant: ten fictional mice at list 25, ten USB-C-to-USB-A data hubs at 35; A/B floors 23/32, C floors equal list. Simulated P1-P6 compatibility only; no charging/display claims. Included charger is a demo condition flag.

### Required non-tech fixtures

All below are fictional brands/products with synthetic attributes, not web-reviewed real items. Each merchant stocks five new units of every variant; A/B floors = 90% of list, C floors = list. Same policy table applies. Reviews show unavailable. Replace/add a genuine identified non-tech item for the real-source smoke test; never relabel these fixtures as real.

| Variants | Typed attributes and USD list prices |
|---|---|
| C1 / C2: Demo Tee | Brand DemoWear; size system Demo US unisex; navy. C1: M, cotton 100%, garment chest circumference 1000 mm/length 700 mm, 25. C2: L, cotton 60%/polyester 40%, 1080/730 mm, 30. |
| H1 / H2: Demo Shelf | Wood, white, freestanding. H1: width/depth/height 500/300/600 mm, 80. H2: 800/400/800 mm, 100. No load-rating assertion. |
| A1 / A2: Demo Kettle | 1.5 L, 220/160/230 mm. A1: 120 V/60 Hz, NEMA 1-15 US plug, 45. A2: 230 V/50 Hz, CEE 7/16 plug, 40. No safety certification asserted. |

A/B maximum item discount is 10%; C permits 0%. The effective floor can be stricter than that cap. For demo contribution accounting, per-unit synthetic cost = 80% of list and required minimum contribution = 5% of list; shipping cost equals the shipping charge. These assumptions are not retailer benchmarks.

All prices are tax-inclusive fictional prices: tax line zero, "included in demo price." Total = sum(quantity x permitted unit price) + shipping once per bundle. No fee/cross-line discount. Agents choose floor-to-list prices; warranty/returns/shipping remain policy constants.

# 7. Agent roles, offer contract and compute policy

One internal Gemini context per merchant; server binds scope before tool calls. Validate arguments and structured final output. Preserve SDK function-call identity/signature parts. No recursive agents, arbitrary browsing, shell access, or model-selected merchant identity. [S3]

| Role | Allowed operations | Cannot do |
|---|---|---|
| Intake | Propose fields, clarify, explain confirmed request | Confirm requirements or purchase for the user. |
| Merchant | searchOwnCatalog, readOwnInventory/Policy, readApprovedSpecs, evaluateOwnPrice, proposeOffer, respondToChallenge, reviseOffer, abstain | Edit stock/policies; see ceiling/rival floors; reserve/pay. |
| Moderator | Read shared requirements/permitted claims; challenge; classify supported/contradicted/unknown | Purchase, alter hard constraints, rank rhetoric, view floors. |
| Evidence | Extract allowlisted sources, assess applicability, attach claims | Change prices, stock, permissions or approval. |

**Offer:** request/run/version; listing IDs/quantities; integer unit prices; policy/inventory versions; claim/source IDs; proposed fit explanation; private contribution/strategy assumptions; reason or abstention. Self-assigned scores never become authoritative buyer scores. Server recomputes total, arrival and terms, validates constraints and persists immutable accepted versions. One repair after rejection within caps, otherwise abstain. Private-constraint rejection is generic to sellers; no budget headroom.

**Events:** event ID, request/version, run ID, round, server-bound role/merchant, offer version, type, timestamp. Types: proposal, accepted, rejected, challenge, response, revision, price_round_opened, final_bid, price_round_skipped, abstained, closed. Challenge names a claim/concession. Responses cite approved sources or admit uncertainty; changed goods/prices require a validated new offer version.

### Compute policy

Default one round: claims arrive with bids; moderator asks at most one targeted question per merchant; each answers and may correct its product/claim once; moderator summarizes changes. Buyer gets a summary and optional claim/tool-outcome transcript, not hidden reasoning. Skip council with fewer than two qualifying merchants.

Allow one additional council round only for a named material issue, a specific new tool result able to resolve it, and time/call/quota headroom. It is a verification round, not another price auction. The separately gated final-price round has at most one submission per candidate. Log trigger. Stop on resolution, no material change, two rounds, cancellation or a limit. Prefer extra compute for relevant evidence checks over repeated persuasion. Defaults are not a proven cost-quality optimum.

Server-configurable caps: three concurrent merchants; two active candidate offers each, with immutable revisions; three tool/model cycles per initial agent; 28 model calls/run including final-price work; 36 only with the extra verification round; 40 tool calls/run; 8,192 output tokens/response; 75-second matching/council, 30-second final-price window, 60-second research, 210-second overall run. Intake is outside the run but capped at three grouped clarifications. One transient retry counts toward caps. Track tokens, elapsed time and provider cost when available. Operator sets provider spending limits before public use; stop on configured quota exhaustion. No promised hard dollar cap without provider support.

Start with gemini-3.6-flash, medium thinking; model/limits are backend settings. Truncated/invalid JSON is rejected. Increase caps only through configuration after measuring cold/warm runs; extra compute cannot bypass policy, identity, approval or time guards.

# 8. Merchant incentives and variable-price policy

Think of a merchant choosing whether to offer a small negotiated concession to win a particular customer. The goal is a profitable completed sale, not the deepest possible discount. A suitable item may win at list price through fit, availability, warranty or delivery.

**Merchant objective:** expected contribution = estimated probability of winning and completing the sale x contribution if sold, less incremental quotation/agent cost. Contribution includes goods cost, fulfillment, payment/platform costs and an allowance for expected after-sales expense where provided. It is not revenue and is not full accounting profit.

The demo has no historical win/conversion data. Agent-estimated probabilities and synthetic costs must be labeled assumptions; do not claim calibrated predictions, truthful equilibrium or a globally optimal price. Record considered prices, margin checks and the strategic rationale privately. An agent can hold price or abstain.

### Enforceable pricing contract

Per listing: reference/list price snapshot, operator floor, maximum discount in basis points (1,000 bps = 10%), provided/synthetic unit cost and minimum contribution. Effective minimum unit price is the maximum of: operator floor; reference price x (1 - discount cap), rounded upward to cents; and unit cost + required minimum contribution. A reference price must be a legitimate current baseline; do not inflate it to manufacture savings.

The agent may propose prices between effective floor and reference price. The server validates cents, quantities, current policy and complete delivered total. Shipping/returns/warranty remain the demo constants; no hidden fee can recover a nominal discount. Policy caps are per product/merchant, never inferred by the model. A/B seed caps are 10%; C is 0%; existing higher operator floors remain binding.

**Important economic limit:** with a $1,000 selling price and $900 relevant cost, contribution is $100. A 5% price cut leaves $50; 10% leaves zero, before any omitted cost. Thus 5-10% can be material. Do not promise every category can afford it. The synthetic seed cost assumptions are demonstration inputs, not estimates of laptop or clothing margins.

### Price disclosure and fairness

- Private maximum: the platform checks affordability without inserting it into merchant prompts or pricing tools.
- Optional shared target: a buyer can explicitly authorize "try for $950." It informs negotiation but is not a promise to buy or permission to charge that amount automatically.
- No inferred wealth/profile pricing: do not use browsing history, identity, demographics or the private ceiling to estimate willingness to pay. The original prompt is private; merchant context is a sanitized projection.
- Different quotes can follow different times, stock, permitted competition or explicit targets. Explain the mechanism. Do not claim identical buyers always receive identical prices, or that stating a true budget is a dominant strategy.

The FTC's study describes personal-data-driven individualized pricing; it is context for the trust design, not a finding about this prototype or a legal conclusion. [R5]

**Why join:** qualified incremental demand, less quotation work, selective concessions, suitable stock movement and competition on service. These are hypotheses to test after fees/returns. Subscription tools or a transparent completed-order fee are possible later; the demo charges no fee. Paid placement cannot influence neutral fit or undisclosed ranking.

# 9. Bidding mechanism and council sequence

**Recommended mechanism:** private initial quotations, a bounded claim council and an optional sealed final-price round. The buyer chooses a final offer and pays its quoted amount in the demo/supported sandbox flow. Because goods and terms differ and the buyer retains choice, this is a competitive quotation marketplace, not a textbook auction with guaranteed truthful revelation. Reverse-auction theory is a useful reference, not a proof of optimal outcomes here. [R6]

1. Freeze the common confirmed requirements, profile versions and disclosed target. Invite three matching demo merchants. Give equal material clarification; private floors, costs, ceilings and raw personal prompts stay scoped.

2. Each merchant independently searches its stock, checks policy, creates up to two candidates, evaluates permitted prices and submits its initial product/price/fit case. Validate technical/commercial facts. Keep policy-valid but over-budget candidates internal and nonselectable: a concession may later make them affordable.

3. The moderator challenges at most one material claim per merchant. Responses cite approved specifications/evidence or concede uncertainty. A correction creates a new version. Debate cannot vote a false hard fact into eligibility. Web research at the end remains independent of this merchant advocacy.

4. The coordinator decides whether final repricing adds value. Require at least two actual independent policy-valid merchants, and either a permitted price change can improve the price order within an equal-fit tier, the buyer requested negotiation, or no current price is affordable but a merchant's permitted floor could be. Log the gate without exposing private ceilings. Otherwise skip.

5. Each participating merchant receives its own validated case and limited truthful feedback: e.g., "your delivered price trails a comparable offer." Do not share rival floors, costs, identity-private strategy or future submissions. Compare equivalent variants/condition/terms where possible; identify material differences rather than pretending all offers are identical.

6. Open a server deadline, at most 30 seconds and inside total run limits. Each candidate has at most one final commercial submission. Store revisions privately until close. Accept lower delivered prices or unchanged prices with expressly permitted improvements; reject price increases, silent reductions in value, or policy violations. Product substitutions are visible and revalidated.

7. Close on the deadline or once all responses arrive. A merchant can hold or abstain; absent/invalid revisions retain its latest valid offer. No endless penny undercutting or reopening because a merchant loses. Revalidate, score, research the shortlist and present actual outcomes.

**One-seller case:** skip the competitive auction narrative. If the buyer explicitly requested a target, permit one policy-bound counteroffer labeled single-seller negotiation; never invent a rival. No eligible stock/floor means an honest no-match, not pressure to break policy.

**Commercial revision safety:** bind round, request/candidate/policy versions and server merchant scope. Logical submission keys deduplicate retries; mismatched replay fails. A completed price round cannot accept late bids. More council compute is for unresolved verification, not extra commercial rounds. If the coordinator fails, return safe current offers with a partial/round-skipped state.

**Merchant intelligence:** the agent should compare contribution at candidate prices, consider whether another concession plausibly changes its chance of selection, and explain why it stopped. evaluateOwnPrice returns deterministic contribution/floor checks; any win-likelihood estimate is an explicitly unvalidated assumption. The server enforces authority even when a model's strategy is poor.

### Worked mock run

Buyer prompt: "I need a new MacBook Air with at least 16 GB memory and 512 GB storage, delivered in San Francisco within five days, under $1,000 total." The assistant confirms that $1,000 is private, quantity one, no extras and no additional soft preferences. These proposed prices are an illustrative trace, not guaranteed agent outputs.

| Merchant | Initial -> final delivered offer | Validation and meaning |
|---|---|---|
| A / P3 | $980 item + $15 shipping = $995 -> $960 + $15 = $975 | Final unit price exceeds its $950 floor and stays inside its 10% cap. $20 improvement from initial delivered offer. |
| B / P3 | $1,000 with free shipping -> $970 | Above its $965 floor; $30 initial-to-final improvement. Final item discount against $1,020 list is about 4.90%. |
| C / P4 | $1,090 -> holds $1,090 | Policy permits no discount. It remains nonselectable under the private ceiling, despite its higher-memory variant. No fictitious concession. |

With no scored preferences, B sorts before A by final delivered price; buyer still chooses. Research P3 once for both sellers, not as two independent product populations. P4 owner evidence is unnecessary if it cannot qualify. Merchant warranty/returns stay separate from product reviews. Selection revalidates and holds the chosen listing before approval; the trace ends with labeled simulated execution.

A non-tech check uses "navy DemoWear M, 100% cotton": C1 can match, C2 cannot. The same commercial engine runs, but the fictional shirt has unavailable owner sentiment; never attach a real brand's reviews to it.

## 10. Neutral fit assessment, arguments and shortlist ranking

The merchant makes the sales case; the platform checks it. A seller's "95/100 fit" is a claim, not a ranking input. The buyer sees both the merchant's argument and an independently calculated requirement breakdown.

**Hard eligibility first:** category/variant/condition, quantities, compatibility, required properties, delivery, commercial authority and private all-in ceiling must pass. Unknown hard facts fail. Policy-valid over-budget proposals can negotiate internally but never appear as selectable qualifying results. A great review score cannot compensate for the wrong size or a missed deadline.

**Default measurable preference coverage:** for each confirmed soft preference that uses a supported typed comparator, match=1 if supported and satisfied, otherwise 0. Use weight 1 unless the buyer confirms a top priority, weight 2. Coverage = 100 x sum(weight x match) / sum(weights). Show matched, unmet and unknown components; an unknown contributes zero and retains its label. With no scorable soft preferences, show no numeric fit score. This is preference coverage, not a probability of satisfaction or an objective quality score.

Subjective language such as "stylish" is explained qualitatively until grounded in a supported preference. The intake assistant proposes priorities in ordinary language; the buyer confirms them. Do not invent weights from a hidden personality profile. Confirmed scoring/profile versions are fixed for the run.

**Default order, a v4 change:** Best fit. With F=coverage descending, P=delivered cents ascending, D=arrival ascending, W=warranty descending and I=stable offer ID ascending, exact keys are Best fit (F,P,D,W,I), Cheapest (P,F,D,W,I), Soonest (D,F,P,W,I), Warranty (W,F,P,D,I). When no score is defined, omit F. Implement these keys in shared server code, never generated prose.

**Shortlist:** maximum two active candidates/merchant. Accepted revisions supersede their parent; history remains. Take the best qualifying offer from each participating merchant, fill remaining positions by selected

order, maximum five, then display that shortlist in the selected order. Explain diversity selection; do not imply these are necessarily the five cheapest offers in the entire pool. Same variant from two merchants is allowed and labeled.

**Evidence consequences:** sentiment itself is advisory. An applicable authoritative fact can invalidate a hard requirement or update a supported field; revalidate/rescore rather than silently editing the merchant snapshot. Research a replacement reserve identity once, at most one refill pass and five primary identities total. Do not silently relax requirements, restart bidding or recycle stale claims.

**Card explanation:** "Meets all 6 required checks; 2 of 3 supported preferences matched, one unknown. Merchant argues the longer warranty justifies $20 more. Owner reports are mixed on fan noise." Each factual fragment must reference a validated attribute, commercial term or evidence record. These labels must remain separable.

**No automatic winner:** the buyer may choose another qualifying offer. Only exact selected terms, a valid reservation and explicit approval authorize execution. The system must not auto-buy because a fit score or negotiated price crosses a threshold.

## 11. Real evidence and final validation

Research distinct finalist product variants after bidding, regardless of category. Merchant-selected sources are inputs, not the neutral corpus. Independently acquire manufacturer/brand facts and owner/test reports. Preflight laptops plus one non-tech product. Never fabricate quotes or transfer reviews from an unrelated product to a fictional item.

**Acquisition:** start an Apify Actor asynchronously; persist run ID/input digest; poll server-side at 2, 5, 10, then 10-second intervals. Verify expected run/status before reading its dataset. Webhooks can wake polling but cannot supply trusted dataset IDs or approval. Abort overdue runs where supported; late data is stale research, never advancing an obsolete request. [S4]

**Bounds:** eight URLs/forty excerpts per primary product variant; five primary variants including refill. Related lines use existing evidence or show not researched; do not imply full bundle coverage. Allowlisted HTTPS sources; no private-network URLs or buyer crawl targets. Deduplicate URL/content/source family. Reuse matching real evidence within 24 hours with timestamp. Cache never establishes stock, price or unit condition. Pin Actor build/input schema. [S4]

| Applicability | Can support | Cannot establish |
|---|---|---|
| Family/model | Related owner themes and family capabilities | Exact size, material, configuration, region or unit health. |
| Exact variant | Explicit attributes and applicable tested performance | Another variant's fit/results or current unit condition. |
| Demo unit/listing | Synthetic diagnostic, stock, price and terms | Real ownership, certified refurbishment or real availability. |

Evidence record: identity/applicability, URL/source ID, type, source date or unknown, retrieval time, short supporting excerpt, claim, Actor/run/build, duplicate group, ownership-label provenance and verification state. Display brief quotations and source links; minimize personal identifiers. Purge/anonymize completed research/requests after seven days; retain unresolved approval/execution/allocation records until reconciliation, then apply retention. No raw full-page retention by default. A demo reset cannot erase unresolved sandbox-attempt evidence.

**Display:** separate neutral fit, merchant sales argument, commercial offer, independent tests, owner reports and demo condition. Show positives, concerns, disagreement, count, freshness and gaps. A claimed owner is not independently verified; attribute site verified-purchase labels to that site. No artificial consensus probability, fake sentiment average or guarantee.

**States:** pending, complete, partial, unavailable, cached. Manufacturer-only says "owner evidence unavailable." Blocked/empty sources can use a previously collected real snapshot with original timestamp and cached label; otherwise show unavailable and source links. Advisory evidence failure does not prevent selection of a fully validated demo offer. Unknown mandatory facts still block eligibility.

**Category caveats:** clothing fit reports retain purchased size/measurement context; appliance performance retains region/test basis; household compatibility requires exact dimensions/interface. Mixed-variant reviews stay family-level. Product reviews cannot prove a refurbished unit's condition, seller reliability or counterfeit status.

**Contradictions:** authoritative facts failing a hard constraint invalidate the offer and trigger bounded refill. Conflicting authoritative facts exclude offers dependent on them until reviewed; a lone allegation/model summary cannot overwrite specs. Web text cannot grant permissions, alter policy, expose secrets or pay.

**Effect on the decision:** sentiment supplies contextual positives/concerns with citations; it does not silently change the fit formula. A new authoritative fact can invalidate a hard constraint or update a supported soft field, triggering revalidation/rescoring. Do not reopen price bargaining because a merchant dislikes the evidence. After finalization, a better offer requires a new run and renewed buyer consent.

## 12. Data, function surface and permissions

Shared runtime-validated TypeScript contracts drive Convex, test fixtures and UI; no parallel handwritten payload definitions. Opaque IDs, integer USD cents, line quantities 1-10, UTC milliseconds; immutable records carry createdAt/schemaVersion. Pass IDs between steps; store source output separately.

| Record | Minimum contract |
|---|---|
| Merchant / membership | demo=true, enabled, name; server-owned userId -> merchantId. |
| CategoryProfile / Product / Variant | Versioned typed fields/operators/units, clarification/compatibility/evidence rules; family identity; exact variant attributes and provenance. Unknown manufacturer IDs remain null. |
| Listing | Variant ID; merchant, condition/unit ID, quantity, list/floor, diagnostics, included items, version. Never collapse all sizes/colors into one stock bucket. |
| Policy / PricingPolicy | Merchant/version, allowed concessions, reference price, discount cap bps, operator floor, synthetic or provided cost and minimum contribution, shipping/terms, effective time. Private cost fields never enter buyer views. |
| RequestVersion / Run | Owner, raw prompt/requirement provenance, confirmed constraints, private ceiling, optional consented shared target, priorities/sort, profile versions; stage, budgets, deadline, config and latest run ID. |
| OfferVersion / CouncilEvent | Request/run/merchant/version, immutable lines/snapshots, terms/expiry; policyValid, requirementsValidExceptBudget, affordable, selectable and reasons; claim/source/event refs. Selectable requires all checks, current version/terms and no unresolved hard fact. |
| FitAssessment / AuctionRound | Versioned score breakdown/coverage; round mode, participant snapshot, deadline, per-candidate final submission key, feedback projection and close reason. |

| Record | Minimum contract |
|---|---|
| ResearchJob / Evidence | Identity/input digest, expected provider run/build, status; provenance/applicability. |
| FirmQuote / Reservation | Request/offer versions, commercial snapshot, hash, expiry; listing allocations, owner, active/locked/released/sold. |
| Approval / Execution / Order | Quote hash, owner/time, attempt key, mode, operation, result/unknown, receipt, mock fulfillment label. |
| ProcessedEvent / Audit | Logical key, payload digest/prior result; timestamps, rejections, token/cost status; no secrets/private prompts. |

### Public functions: authenticated and scoped

- saveDraft/confirmRequest/reviseRequest: owner; validate fields; confirmed revision immutable. Changes cancel obsolete jobs/nonlocked holds; executing requests cannot change.
- startRun/cancelRun/getBuyerView: owner; one active run/request, three active requests/session, five starts/minute. Reload resumes stored progress.
- getMerchantView/updateOwnListing/publishOwnPolicy/pauseOwnMerchant: server membership. Stock edits cannot erase sold/held allocation. Pause blocks new bids/selections, not locked-attempt reconciliation.
- getBuyerView includes only safe score/claim/auction summaries. selectAndReserve: owner/current offer/version; transactional recheck/hold. approveAndExecute: owner/exact quote hash/unexpired hold/idempotency key.

**Internal only:** dispatchAgent, validateAndStoreOffer, assessFit, openPriceRound, submitFinalBid, finishResearch, closeRound, expireHolds, executeAdapter, reconcileExecution and resetDemo. Model merchantId grants nothing. Same key + same payload returns prior result; mismatched replay fails. Every completion checks active version, expected stage/run and transition.

**Auth:** Convex Auth Anonymous for buyers, Password for privately provisioned operators. Create three accounts, grant memberships internally, then reject password signUp server-side. Never trust browser role/email fields. Buyers read own runs; merchants read own policies/bids and permitted council projection, never ceiling/rival floors. Separate buyer/operator browser contexts. Guest cross-device recovery is out of scope. [S2]

**Errors:** code, safe message, retryable, correlationId. Codes: INVALID_INPUT, FORBIDDEN, STALE_VERSION, NO_MATCH, POLICY_REJECTED, EXPIRED, STOCK_UNAVAILABLE, PROVIDER_UNAVAILABLE, LIMIT_REACHED, EXECUTION_UNKNOWN. Seller errors omit private values. Operator changes invalidate unapproved stale quotes; an executing approved snapshot completes under its frozen terms.

Budget/target cents must be positive safe integers; a shared target cannot exceed the confirmed maximum. Discount bps are integers 0-10,000, with stricter configured caps. Price-round eligibility requires policyValid and requirementsValidExceptBudget, not selectable; otherwise an initially unaffordable candidate could never negotiate into range. Feedback is bounded, not a claim of preventing every inference about hidden limits.

## 13. Lifecycle, reservations and execution safety

Request: draft -> bidding -> council -> negotiating (optional) -> researching -> results -> reserved -> executing -> complete. Alternatives: no_match, cancelled, failed. Partial/unavailable evidence annotates results. Persist state; browser/model memory is not authoritative.

On stage/overall deadline, close outstanding jobs. Continue with already validated offers and truthful partial evidence; if none exist, use no_match for confirmed inventory mismatch or retryable failed for provider failure. Late completions cannot reopen the run. Never manufacture agent output when generation failed.

| Transition | Guard and outcome |
|---|---|
| Confirm/start | Owner confirms required fields; create version/run; schedule workflow. Edit invalidates old work and releases active nonlocked holds. |
| Bids -> results | Validate current offers; apply council and optional sealed final bids; freeze current comparison pool; research/refill within limits. |
| Select -> reserved | Recompute current policy/listing; reject paused/unavailable/noncompliant; atomically create immutable firm quote and allocations. |
| Approve -> executing | Check owner/hash/versions/server time; create one approval/attempt and set execution_locked before provider calls. |
| Success | Convert locked allocations to demo sold exactly once; save truthful mode-specific receipt. |
| Definite decline/failure | Record terminal attempt; release once; return buyer to selection. Unknown is not a definite failure. |

**Timers:** offer validity starts at each accepted version, 15 minutes; final-price deadline is server-owned, at most 30 seconds and within the run budget; no hold during research. Expired selection requires full revalidation, not automatic rejection forever: issue a renewed firm quote and display any changed terms. Firm quote/hold valid three minutes; server checks time even if cleanup is late. Successful replacement atomically releases old hold; failed replacement keeps the old valid hold.

Available = initial quantity minus sold and unexpired active or execution-locked allocations. Expired active holds do not count even before cleanup. Unique refurbished units have quantity one. Release/expiry/sale is idempotent. Never release an execution lock just because its former three-minute hold elapsed.

### Payment adapter and truthful results

Default PAYMENT_MODE=simulation. No real card collection/live-money capability. Adapter input is server execution ID + approved quote; output is succeeded, declined, pending or unknown, actual operation and provider reference. Privileged simulation fixtures: success, declined, unknown_then_success. Success says "Demo order completed; no payment taken"; delivery remains fictional.

Enable Visa sandbox only after exact API/test mapping, amount/merchant semantics, duplicate protection and status lookup pass a smoke test. Disclose test-merchant mapping. An unrelated fixed test amount is not proof the chosen quote was paid. Tokenization-only says "Visa sandbox credential test"; order completion stays explicitly simulated. No invented endpoints/settlement claims. [S5]

Timeout -> EXECUTION_UNKNOWN with allocation lock retained. Reconcile the same attempt/reference; block another purchase for that request until resolved. If safe status/idempotency handling is unavailable, disable that external path and use simulation. Disable cancellation/edits during executing/unknown. Internal demo reset labels fictional-state reset; it cannot imply a provider refund.

A final-price timeout retains the latest valid initial/corrected offer for merchants that did not improve; it does not fabricate a concession. Expired or invalid offers remain nonselectable. Frozen comparison snapshots distinguish initial, revised and final terms.

# 14. Implementation Plan: architecture and setup

**Runtime:** React/Vite SPA -> Convex functions/Workflow -> Gemini actions and Apify jobs -> stored results -> approved quote -> adapter. Vultr serves built SPA over HTTPS with client-route fallback. Convex hosts backend; Apify hosts extraction. No permanent extra worker. [S1, S4]

Convex mutations own state/stock transactions; actions own external calls; Workflow persists orchestration/retries. Use official @google/genai behind an adapter, pin SDK/lockfile and preserve function-call parts. Source registry pins Actor builds/input mappings. Optional Product Evidence Pack Actor normalizes existing crawlers; no new crawler required. No network inside transactions. [S1-S4]

### Configuration

Browser: public VITE_CONVEX_URL/demo labels only; VITE variables are public. Backend: GEMINI_API_KEY, GEMINI_MODEL, APIFY_TOKEN, source registry, PAYMENT_MODE, optional Visa test credentials/mapping and Auth signing settings. Server constants own limits/TTLs/region. Keep credentials and provisioning gates out of client/public source. [S1-S4]

Start from the official Convex React/Vite + Auth scaffold. Seed catalog/policies and privately provision operators; disable public password signup. Default inventory mode is mock. Never assume credits or generate credentials in source.

### Release sequence

1. Freeze contracts/profile definitions and Figma states; seed fixtures/auth; prove owner and merchant isolation.

2. Test real Gemini tools and Apify sources for a laptop and non-tech item; pin working versions. Failures become honest cache/unavailable states.

3. Connect one full offer -> hold -> approval -> simulation path, then council/evidence/refill; exercise each category without a backend fork.

4. Configure deployment Auth/Convex separately; build dist for HTTPS Vultr hosting with index.html fallback. Vite preview is not production. Test nested refresh/fresh sessions.

5. Run acceptance cases, gated Visa test and submission proof. No deployment/spend/publication performed by this planning task.

Use the laptop pitch plus a short non-tech run; clearly distinguish synthetic fixtures, live/cached evidence and simulation.

**Module boundaries:** shared contracts/category profiles; buyer intake/projection; merchant inventory/policy; scoped agent adapter; pricing evaluator/auction coordinator; fit validator; evidence identity/acquisition; reservation/execution; safe UI queries. External services sit behind adapters so deterministic fixtures can test orchestration without pretending that fixture output came from a real model or crawl.

# 15. Figma design, prompt UX and frontend handoff

Two teammates own Figma design and the UI handoff. A owns buyer intake, requirements, progress, comparison/evidence and checkout/result; B owns merchant catalog/policy, bid/council views and shared components. This ownership split is a proposed working default. Figma work must become a connected frontend; a clickable design alone does not execute the marketplace.

Freeze the shared contracts first: CategoryProfile, RequestDraft, BuyerRunView, OfferCard, EvidencePanel, FirmQuote, ExecutionView, MerchantView and ApiError, with schema-defined permission-safe fields. Backend owns totals, eligibility, rank, timestamps, modes and allowedActions; UI renders them. The prompt is the

primary input; profile fields drive assistant clarification, optional detail chips and comparison rows, never a mandatory specification form or laptop-only columns. Fixture payloads cover all four category families. Figma may supply components/tokens; server data remains authoritative.

Design/handoff must cover responsive layouts, keyboard/focus, form validation, extraction ambiguity, target-sharing consent, matching/council/repricing progress, loading/empty/error, stale offers, partial evidence, expired hold, payment pending/unknown and success/decline. Persist IDs across reload; use subscriptions for progress. Council detail shows claims/outcomes, not private reasoning. A/B own frontend wiring by default, C/D supply tested contracts and adapters; no unassigned conversion from design to code.

**Figma tracks:** supplied and current event listings include Best Use Case of Figma, Best Design, Most Creative, plus two Figma Agent categories. Preserve the design file, prototype, iterations and live implementation evidence. Agent categories require actual demonstrated Agent use and rule confirmation; ordinary Figma use does not establish that. Figma is a development/design track, not an added runtime dependency. [S6]

**Buyer-facing language:** use "Tell us what you need," "Here is what we understood," "Merchants are preparing offers," "An offer improved by $X," and "What owners report." Do not surface SDK names, model internals or policy JSON in ordinary shopping screens. A separate operator/debug view may show validated tool outcomes.

**Required card content:** exact product/variant/condition, neutral fit explanation and coverage, merchant argument, initial/final delivered total and true discount basis, delivery, warranty/returns, source-backed sentiment/gaps, expiry and selection action. Only show "best price" or "best fit" when its published comparison rule supports the label.

# 16. Sponsor strategy: core targets and listed prizes

Use technologies because they perform real project work, then claim only the tracks supported by evidence. The prize values below are transcribed from the supplied event snapshot; the current official event page was checked. Some headings say cash while descriptions name hardware/gift cards. Report both where they conflict; do not sum them into expected winnings. Entry caps, award stacking and full sponsor rules remain unconfirmed. [S6]

| Target track | Listed prize | Meaningful role and proof to preserve |
|---|---|---|
| [Visa] Best use of Reimagining Shopping sponsored by Visa | $5,000 heading | Primary narrative: consumer intent becomes merchant competition, explainable offers and approved checkout. Show every rubric dimension; attempt actual sandbox operation, accurately label simulation. API mandate/access not established from listing. |
| [Apify] 1st/2nd/3rd/4th/5th Place Best use of Apify | $2,000 / $1,000 / $800 / $400 / $400 headings | Real final-product evidence acquisition and normalization. Preserve Actor/run IDs, source provenance and the working extraction path. Places are alternatives within one competition, not five separate integrations. |
| [Apify] Best Actor sponsored by Apify | $400 heading | Conditional extension: reusable Product Evidence Pack Actor consumes source inputs and emits identity-matched evidence. Confirm custom Actor/publication/separate submission rules before claiming qualification. |
| [Convex] Best use of Convex | $1,308 heading; description: Teenage Engineering EP-133 K.O. II plus Waveshare ESP32-S3 touch display | Authoritative requests, isolated policies, bid versions, sealed rounds, reactive progress and atomic mock reservations. Demonstrate durable recovery and server checks; simple database presence alone is a weak story. |
| [MLH] Best Use of Gemini API | MLH Swag Kits | Real Gemini API for prompt interpretation, merchant strategies/arguments and bounded council/evidence synthesis. Preserve working tool calls; fixtures alone do not establish API use. |
| [MLH] Best Use of Vultr | M5Stack Official Tab5 | Host the actual built frontend over HTTPS. Convex remains backend; no fake extra worker needed. Preserve hosting configuration and live endpoint. Credits/access unconfirmed. |
| [Figma] Best Use Case of Figma | $1,000 heading; description: Amazon gift cards and Early Career Recruiting session | Core design effort by A/B: prompt confirmation, cross-category comparison, evidence and merchant controls. Preserve design/prototype/iterations plus connected implementation. Exact judging proof remains to confirm. |
| [Figma] Best Design; [Figma] Most Creative | Each $1,000 heading; same gift-card/session descriptions | Candidate design categories for understandable commercial negotiation and trustworthy comparison. Confirm multiple Figma entries are allowed and substantive criteria; no eligibility guarantee. |
| [Figma] Best Use Case of Figma Agent; [Figma] Most Creative Use Case of Figma Agent | Each $1,000 heading; same gift-card/session descriptions | Conditional only if the teammates actually use and demonstrate Figma Agent. Ordinary Figma design alone is insufficient evidence of Agent use. Access/rules must be confirmed. |

**Core stack logic:** Gemini reasons; Convex enforces/coordinates; Apify acquires evidence; Visa anchors responsible commerce; Figma shapes the experience; Vultr hosts the frontend. These roles complement each other. Avoid duplicate agent frameworks or databases.

**Submission evidence:** save a short integration log for each claimed sponsor: feature, code/config location, one reproducible demonstration, real-versus-simulated mode, account/source limitations and required links. Do not claim that a logo, unused SDK or development assistant automatically satisfies a runtime-use track.

## 17. Sponsor strategy: conditional additions

These tracks have plausible uses but are not baseline dependencies. Prize details follow the supplied snapshot; eligibility and access are unresolved. Add a track only after the complete core flow and acceptance gates work. [S6]

| Track | Listed prize | Fit, cost and decision |
|---|---|---|
| [Factory] Best Use of the Factory Droid SDK | $2,000 cash, credits and swag | A narrow merchant connector/normalization assistant using the actual SDK could help later. Developer use alone does not establish SDK integration. Deferred with mock inventory; do not add a second autonomous council harness. |
| [Cartesia] Best Use of Cartesia | $1,516 heading; description Meta glasses, TBD | Voice intake could complement the prompt for accessibility, with editable transcript and written confirmation. Deferred; narrated bidding adds little. Text remains primary. |
| [Elastic] 1st/2nd Place Best use of the Elastic Agent Builder on a Serverless instance | $1,200 / $1,436 headings; descriptions Meta Quest 3S / Bose QuietComfort | A specialist evidence/retrieval agent is useful at scale. Named track requires Agent Builder on Serverless; ordinary search is not proof. Deferred unless selected to replace an existing research component. |
| [Creao AI] 1st Place Best AI-First Disruptor; Best One-Person Company; Best Team Empowerment | Each $8,760 heading; body $3,000 cash plus CREAO subscription, mentorship and other opportunities | Merchant automation fits conceptually. Required CREAO use/definitions unknown; avoid duplicating the core platform for a headline prize. One-person business beneficiary does not override hackathon team rules. |
| [Moss] Best Use of Moss | $996 heading; AirPods Max/Pro description | Potential private catalog/evidence semantic retrieval. Verify sponsor identity and exact rules first. Redundant for a small typed catalog; not selected. |
| [Microsoft] 1st Place Built for the Real World; 2nd Place Microsoft Customer Impact Award | $3,800 / $0 headings; Surface laptops, presentation and mentorship descriptions | Potential outcome fit, stronger with real merchant pilot. No qualifying Microsoft product requirement established from supplied copy; do not invent one or claim eligibility. |
| [Hyphenate] 1st/2nd/3rd Place The Hyphenated Enterprise: Autonomous Systems for Finance | $8,000 / $5,000 / $3,000 cash; expedited interview process for top five teams described | Margin governance and reconciliation are adjacent, but shopping debate alone is not an enterprise-finance workflow. Deferred; do not pivot away from Visa for prize size. |
| [Coinbase] Best Use of x402 by Coinbase | $2,000 heading; body $500 USDC/member/track, two tracks/up to $4,000 total | An agent-paid metered evidence API is plausible later, separate from buyer Visa checkout. Wallet funding adds complexity. Unnamed second track must be clarified; no invented category. |
| [Armadin] Best Security Hack | $996 heading; Apple headphones, possible hacking tools | Consider only with substantial adversarial security evidence and confirmed brief. Having sensible permissions does not by itself establish track qualification. |

| Track | Listed prize | Fit, cost and decision |
|---|---|---|
| [MLH] Best Domain Name from GoDaddy Registry | Digital Gift Card | Low incremental work if an eligible relevant domain is registered. Confirm route/suffix requirements; domain purchase is not performed by this planning task. |
| [MLH] Commit Fellowship Prize Category | Interview and The Lean Startup | Potential continuation path for a working prototype and real problem. Listed brief requires opt-in and supplemental form. Not automatic fellowship acceptance. |

**Decision rule:** one useful new capability, one bounded integration, demonstrable real use and no duplicated source of truth. If it fails that rule or delays bidding/evidence/approval reliability, omit it from the submission claims.

# 18. Sponsor strategy: alternatives, exclusions and discrepancies

| Track / technology | Listed prize | Reason not selected |
|---|---|---|
| Fetch AI / [Fetch.ai] Best Use, places 1-3 | Devpost snapshot $1,500/$1,000/$500; sponsor hackpack $1,250/$750/$500 plus recognition prizes | User prefers own UI. Official pack requires Agentverse, ACP, ASI:One discovery and the entire primary workflow inside ASI:One, plus public runnable repo and additional submission. Thus excluded under the confirmed UI decision. Prize discrepancy is recorded, not silently resolved. [S6] |
| Spacetime | No named track/amount in supplied prize list | Official event blog announces a special prize but supplies no amount here. It could replace Convex for shared real-time state; adding it as a second authoritative database is unjustified. Confirm actual brief before reconsidering. [S11] |
| [Bitdeer AI] Best Use of Bitdeer AI Model Studio, places 1-2 | $500/$200 headings; body credits and gaming mice/LEGO | Alternative model hosting if genuinely needed; no baseline reason for another provider. No track claim without actual Model Studio use. |
| [MLH] Best Use of Snowflake API | Arduino Student Kit | Alternative model/analytics platform. Current small demo already has Gemini/Convex; no additional necessary role. |
| [MLH] Best Use of Tiger Data | Stream Deck Mini | Alternative primary database or meaningful later analytics. Do not copy the same bid state into two backends for a sponsor name. |
| [MLH] Best Use of Solana | Assorted prizes | No current decentralized requirement. Adding a token transfer does not improve merchant pricing or evidence reliability. |
| [Capital One] Best Use of Nessie | $1,200 heading; $300 Giftogram/member description | Mock banking could inform a budgeting extension but cannot establish actual funds, reserve inventory or execute Visa checkout. Redundant simulation now. |
| [CRS] Best Overall CRS Hack; Best Financial AI Agent; Best Identity / Fraud Innovation; Most Unexpected Use of CRS | $1,000/$500 headings; latter two interview opportunities | Consumer credit/identity data does not prove product quality or merchant stock. No credit pulls or financing feature in the core shopping demo. |
| [Meta] Meta/Facebook: Bringing People Closer Together with AI | Next-round invitation, travel/accommodation subject to eligibility, swag | Buyer-seller commerce is not automatically a people-connection innovation. No social pivot solely for this category. |
| [SpaceXSI] Make it Legendary with SpaceXSI | $400 heading; keyboards/raffle described | Supplied brief concerns space data and requires Cursor. Off-theme for current marketplace. |
| [Notability] Best Use of Notability | $396 heading; Pro and merchandise | No necessary product role established. Using a tool incidentally is not a reason to claim the track. |

**Known discrepancies:** prize headings versus hardware/gift/credit descriptions; Fetch sponsor amounts versus Devpost; Spacetime sponsor announcement without listed category/amount; Coinbase body mentioning two tracks but only one named heading. Resolve with organizers; do not assume all are cash or simultaneously winnable.

**Unconfirmed rule:** no verified maximum sponsor-category count or award-stacking policy was found in prior review. Registration filters do not establish permission. Confirm entry combinations, separate sponsor portals and required artifacts before submission.

# 19. General awards, event requirements and demonstration

The supplied event list also contains general awards. They are quality opportunities, not reasons to add unrelated technologies. Listed rewards below are from the supplied snapshot, not a promise of award value or eligibility. [S6]

| Award | Listed reward / fit |
|---|---|
| Cal Hacks First Overall | $3,600 heading; MacBook Air or Framework laptop/member description. General quality target. |
| Cal Hacks Second Overall / Third Overall | Nintendo Switch 2 / Google Intelligent Eyewear per member. General quality targets. |
| Cal Hacks Most Creative Project | Instax camera plus film/member. Distinctive merchant offer formation and an understandable negotiation story fit conceptually. |
| Cal Hacks Most Technically Advanced Hack | NuPhy Air60 HE/member. Prove scoped tools, versioned offers, policy enforcement, evidence identity and failure recovery. |
| Cal Hacks Hacker's Choice | Logitech MX Master 4/member. A clear short live experience is more persuasive than a wall of agent messages. |
| Cal Hacks Best Beginner Hack | AirPods 4/member; team eligibility unknown. |
| Cal Hacks Best People-First Social Impact Hack | Apple Watch SE 3/member. Conditional on demonstrated user benefit; affordable-looking prices alone are insufficient. |
| Cal Hacks Best Planet-First Social Impact Hack | Parks Pass/member. Refurbishment does not automatically prove reduced environmental impact; requires defensible reuse evidence. |
| Cal Hacks Best Hardware Hack | Bambu Lab 3D printer/member. No current hardware component. |
| Cal Hacks Best Solo Hack | AirPods Pro 3. Not a target for this four-person team; general listing says teams of 2-4, so apparent category inconsistency is not assumed resolved. |

### Event and submission checklist

Official header lists October 25, 2026, 9:00 AM PDT. Body copy uses PST inconsistently and lists venue/check-in requirements; confirm practical timing with organizers. Public requirements include work during the event, a project image, repository, live or recorded demonstration and a three-minute pitch plus two-minute questions. No external submission or publication has been performed here. [S6]

**Three-minute story, suggested:** 0:00-0:30 buyer prompt and concise confirmed brief; 0:30-1:15 actual scoped merchant tool activity and contrasting fit cases; 1:15-1:45 one truthful competitive feedback/final concession or justified no-discount; 1:45-2:20 independent sentiment, exact variant caveats and final tradeoffs; 2:20-3:00 selected quote, approval and truthful simulated/sandbox outcome. Keep a recorded fallback with the same disclosed modes.

Also preserve a short non-tech run and an execution-unknown recovery clip. The primary demo can be a laptop without implying a laptop-only product. Show mock labels prominently, current commercial reference, initial/final prices, source timestamps and actual merchant participation. Never portray three fictional catalogs as three onboarded real businesses.

**Evidence pack:** Figma file/prototype, screenshot, repository/run instructions, provider feature proofs, source provenance, bounded run metrics, failure outcomes and sponsor-specific supplemental artifacts once verified. Explain which pieces are live, cached, synthetic and simulated.

# 20. Existing products, differentiation and alternatives

Buyer requests, supplier quotes, consumer deal distribution, AI catalog search and autonomous negotiation all have precedents. These comparisons rely on official descriptions, not hands-on testing. Their existence does not prove demand for BuyerUnion or rule out another startup with similar features.

| Product / pattern | Overlap and credible distinction |
|---|---|
| Alibaba.com RFQ | Buyers post product requests and suppliers respond with quotations. Direct overlap with the marketplace mechanism. BuyerUnion proposes consumer-scale, low-friction requests, authorized agent offer formation and independent product evidence. [C1] |
| Amazon Business RFQ | Earlier research documented bulk requirements, supplier offers and checkout. The official page returned 403 during this refresh, so current eligibility/thresholds are not restated as verified. It remains a procurement precedent, not proof of consumer autonomous bidding. [C2] |
| Carwow | Consumer car configuration and retailer offers show a buyer-led vertical model. Current partner materials describe enquiry-funded distribution. BuyerUnion spans manufactured products and proposes constrained agent pricing and evidence rather than only referrals. [C3] |
| Google Direct Offers | Merchant-controlled promotions and AI commerce substantially overlap with personalized offers. BuyerUnion's proposed difference is a persistent buyer request, independent merchant strategies and transparent comparison instead of undisclosed promotional priority. No universal availability is assumed. [C4] |
| Shopify Agentic Storefronts / Catalog | Merchant catalog and checkout distribution into AI channels are becoming infrastructure. These may be future connectors; a shopping chat over catalog data alone is weak differentiation. [C5] |
| Pactum | Enterprise procurement agents run RFQs and negotiate terms. Autonomous negotiation is already a product category. BuyerUnion proposes merchant-side agents for ordinary consumer purchases, with a simple buyer flow. [C6] |

**Best differentiation to demonstrate:** the buyer's ordinary-language intent becomes a common confirmed request; each seller's authorized agent constructs a feasible profitable offer; bounded competition changes a real permitted price/term; neutral checks and independent sentiment explain the final tradeoff; approval binds exact terms through execution.

### Design alternatives considered

- Public live undercutting is dramatic for identical goods but exposes more strategy and encourages repeated price-only competition. Private initial offers plus one final revision are the default.
- Fixed policy-generated quotes are fast and essential as a baseline. Agents must improve matching, explanation or authorized concessions enough to justify their cost.
- A buyer maximum shared with all sellers can simplify targeting but anchors prices and weakens privacy. Keep it private; allow an explicit target instead.
- A debate vote or seller-authored score is easy to manipulate. Use merchant advocacy plus independently computed requirements and published ranking.
- Cross-merchant bundle optimization could improve selection later but adds multiple stock/payment/return responsibilities. Keep each demo bundle with one merchant.

**Defensibility hypothesis:** reliable merchant integrations, trusted policy enforcement, outcome history and useful buyer demand are more durable than model choice or agent count. No unique-first or proven-moat claim is made.

# 21. Trust, platform incentives and commercial risks

The platform supplies seller agents and also helps buyers compare them. This conflict must be managed through separate scopes, transparent evaluation and visible commercial relationships; it is not solved by calling one model neutral.

| Tension / reachable failure | Requirement and residual uncertainty |
|---|---|
| Merchant margin versus buyer discount | Enforce independent floors/caps and permit no concession. Demand may fail at an honest price; do not force a sale. |
| Private budget becomes a price anchor | Sanitize raw prompts, isolate the ceiling and audit merchant inputs. Explicit targets are optional. Buyers may still strategically understate a target; truthfulness is not guaranteed. |
| Shared software sees competitor secrets | Separate merchant contexts/policy access and no cross-merchant pooling of floors or future pricing plans. Platform operators are not cryptographically blind; real deployment needs specialist review of competition/privacy obligations. |
| An agent invents suitability or authority | Typed validators, per-attribute provenance, exact variant/unit distinctions and current policy checks. Unknown hard facts block the offer. |
| Cheapness hides worse terms | Compare full delivered totals and conditions. A final revision cannot silently weaken warranty, returns, compatibility or delivery. |
| Fake competition or fake sentiment | Show actual participant count, real sources, ownership-label provenance and evidence gaps. Mock merchants are not independent real businesses. |
| Lead spam and low buying intent | Bounded relevant invitations, private contact details, request expiry and rate limits. A request is not a purchase commitment; require no deposit in the prototype. |
| Neutral rank versus platform revenue | No paid ranking in the demo. Future fees must be visible and independent of neutral fit. A completed-order fee still creates incentives toward larger orders. |
| Price creep or speculative shipping | Snapshot terms, expire offers, reserve selected stock only and reapprove changes. Demo arrival/tax policy is fictional and visibly so. |
| Limited market liquidity | Validate useful merchants per category/location/deadline, not total registration count. General product scope does not eliminate local supply gaps. |
| Scraped pages inject instructions | Read-only evidence boundary, source allowlist and no authority-bearing tool access. Scraped claims cannot approve, change policies or execute payment. |

**Revenue direction, unresolved:** prefer testing a transparent completed-order charge or paid merchant automation over pay-per-bid. The first aligns revenue with an outcome but raises fee/refund/bypass issues; subscriptions need enough recurring demand; lead fees can reward noise. No monetization is implemented or fee rate assumed in the demo.

**Seller and support roles:** eventual merchants remain identified sellers and fulfillment providers; the platform owns matching, offer presentation and its own support obligations. Do not promise escrow, guaranteed authenticity, delivery insurance or adjudicated refunds without a separately designed service. Keep the selected offer record for support; the prototype ends at an explicitly fictional order.

# 22. Impact, experiments and success measures

The intended outcome is a suitable, understandable and potentially better-priced purchase opportunity that the merchant can honor profitably. A hackathon demo can prove mechanics; it cannot establish actual conversion, loyalty or merchant profitability with fictional inventory.

| Question | Measure / fair comparison |
|---|---|
| Does prompt intake save effort? | Buyer corrections, clarification turns and time to confirmed accurate need versus ordinary filters. Check misunderstanding, not only speed. |
| Does competition improve offers? | Initial-to-final delivered price, terms and contribution on the same validated candidate; proportion improved, held and abstained; zero/one/multiple merchant participation. |
| Are agents better than a rule? | Compare fixed catalog matching, automatic maximum discount, and bounded agent bidding on identical fixture requests and policies. Hold inventories/requirements constant. |
| Do buyers receive value? | Time to qualified offers, comparison effort, task completion, understanding of tradeoffs and equivalent delivered cost. Budget minus paid amount is not savings. |
| Do merchants receive value? | Contribution after permitted concessions and fees, quotation effort, incremental completed sales, returns and repeat participation. Mock results are simulated. |
| Is evidence useful? | Exact-variant coverage, cited/uncited claims, duplicate rate, source freshness, disagreement and correct unknown labels. Number of scraped pages alone is not quality. |
| Is execution dependable? | Reservation contention, stale quote rejections, changed terms, duplicate attempts, unknown reconciliations and correct mode-specific receipts. |
| Is compute justified? | Cold/cached latency, model/tool calls, token/provider cost and material corrections per additional round. Compare one council round against a justified extra verification round. |

### Experiment plan

Create an offline request suite across all category profiles: exact requests, broad use cases, conflicting requirements, tight deadlines, low/private ceilings, disclosed targets, scarce stock and missing facts. Replay the same states under three pricing baselines. Report validity and margin separately from discount depth. Synthetic buyer-choice outcomes test the simulator, not human conversion.

Next, recruit a small consented pilot with real buyer needs and authorized merchant policies. Ask whether sellers would honor the resulting quote after their actual costs and whether buyers would choose it after ordinary alternatives. Test whether discounts create new purchases or merely reduce revenue on purchases that would have happened anyway. No sample size or representative outcome is claimed in advance.

**Discount reporting:** compare the exact item's genuine reference price and final item price, plus the complete delivered totals. For different bundles or terms, describe the difference; do not claim an equivalent saving without a defensible baseline. Never manufacture a reference price or count the private budget as the baseline.

**Go/no-go after a pilot:** poor inventory authority, mostly single-seller coverage, merchant refusal to repeat, larger research effort than ordinary shopping, or negligible improvement over fixed filtering would challenge the business. A technically impressive council is insufficient if the offers are unreliable or economically unattractive.

# 23. Functional and failure acceptance requirements

| Test group | Required expected result |
|---|---|
| Intake / categories | Broad/exact input confirms a profile. Windows/16GB/512GB rejects P6. Cotton 100% + DemoWear M rejects C2; width <=600 mm rejects H2; US 120 V/60 Hz plug rejects A2. No laptop fields required on clothing. Unknown category stays draft. |
| Policy | Below-floor, wrong SKU, false warranty, incompatible accessory, bad cents/quantity or unknown hard fact rejected despite valid JSON. One repair or abstention. |
| Privacy | Buyer A cannot read B; merchant A cannot read/edit B; forged roles fail; ceiling/floors absent from opponent prompts, public data and logs. |
| Council / rank | Three actual isolated tool-using contexts; bounded rounds; rhetoric changes no rank; self-scores cannot override neutral assessment; same snapshot sorts identically; extra round logs a material trigger. |
| Evidence | Wrong configuration/size/region not exact evidence; unknown health fails threshold; synthetic C/H/A fixtures get no borrowed reviews. Laptop and genuine non-tech source retrieval tested; blocked/injected pages stay safe. |
| Version / refill | Edits make old completion stale; factual invalidation refills once within limits; fewer results truthful; no fabricated review/restart. |
| Inventory | Last unit: one hold wins; delayed cleanup cannot permit expired approval; duplicate release restores nothing twice; failed selection swap preserves valid hold. |
| Execution | Double click: one attempt/order; expiry preserves execution lock; unknown_then_success reconciles once; decline releases once; tokenization never claims charge. |
| Recovery / UX | Reload resumes; timeout produces partial result/retryable failure; nested refresh works; modes visible. All Figma handoff states connect to real typed responses; keyboard completion works. |

### New pricing and prompt cases

- A broad prompt leads to a short conversational clarification and confirmed summary; no technical form is required. An ambiguous price remains unconfirmed; merchants never receive raw private ceiling text.
- Same public request/current market with a different private maximum: merchant initial pricing inputs stay unchanged; do not require bit-identical nondeterministic model outputs. Affordability/selection and the bounded negotiation gate may differ; private limits do not become quote anchors. An explicitly shared target is an allowed input.
- A/B accessory bids below 23/32 USD are rejected. A 10%-cap policy rejects larger reductions even if the operator floor is lower. C can win without discount or abstain.
- A policy-valid offer above the private ceiling may remain an internal negotiation candidate; it cannot be selected. Final concessions can make it affordable without revealing the ceiling.
- No genuine rival means no competitive-auction claim. A requested one-seller counteroffer is labeled negotiation. Final bids remain hidden until close; duplicate/mismatched submissions are handled idempotently.
- A seller that does not revise retains its valid earlier offer. A revision that raises delivered price or silently weakens terms is rejected. A new product must pass all constraints and be visibly identified.

- Eligible-only ranking and score components are reproducible. A seller's persuasive prose cannot make unknown hard facts pass. Savings use the same item/condition/quantity/terms/reference snapshot.

# 24. Implementation milestones, ownership and resources

| Milestone | Work split and exit gate |
|---|---|
| M0: foundation | A/B Figma states and UI contracts. C category/catalog/policy/auth contracts + model tool test. D state/holds/payment + source access test. Agree schemas before parallel UI work. |
| M1: thin path | A buyer UI + B merchant UI wired to C valid offer + D hold/approval/simulation. Reject unauthorized transitions. |
| M2: differentiation | C merchant agents/council/profile validation and sealed repricing. D evidence/identity/refill and Workflow. A/B progress/comparison/council/evidence screens; exercise laptop, clothing, household and appliance fixtures. |
| M3: hardening | C policy/privacy/adversarial tests; D lock/reconciliation, Visa gate and source failures; A/B all error/expiry/accessibility states. |
| M4: submission | A/B Figma and demo evidence; C/D integration proof. All repeat runs, video/image/repo, truthful modes and category eligibility. |

With two backend builders, prioritize the shared flow and category tests. If time runs short, cut optional integrations/Actor packaging before category support; retain a laptop and one non-tech end-to-end demo. Candidates: Visa, Apify, Convex, Gemini API, genuine Vultr hosting and Figma; Best Actor/Figma Agent conditional. Fetch AI excluded by UI choice; Spacetime replaces Convex. Elastic/Factory/Cartesia deferred. Compare agents against plain catalog filtering.

**Owners and integration seam:** A buyer Figma/UI, B merchant Figma/UI, C catalog/policy/agents/pricing, D orchestration/evidence/reservations/execution. C/D agree state and pricing contracts together; A/B consume frozen fixtures before keys arrive. Shared changes require updating contracts and fixture snapshots, not separate ad hoc payloads.

**M0 exit:** functioning development auth, all-category seed import, extraction schema and sanitized merchant projection, one model tool call, one real source extraction, initial budget caps, and a connected UI shell. M1 must work before extra sponsors. M2 adds actual differentiation. M3 runs the adversarial suite and records cold/warm timing. M4 preserves the exact demo configuration and proof of each claimed integration.

**Resources:** Convex deployment; Gemini API project/quota; Apify account/Actor access; Figma collaboration access; optional Visa sponsor credentials; Vultr hosting/HTTPS access; source registry; disposable operator accounts. No credentials belong in the document or repo. Provider access, credits, source success and an optimal runtime budget are not assumed.

# 25. Roadmap and path from prototype to real commerce

### Prototype completion

Complete prompt -> common brief -> scoped inventory search -> validated offers -> council -> conditional final bids -> neutral fit and independent sentiment -> reservation -> explicit approval -> truthful result. Exercise all four profile families; a laptop and one genuine non-tech product show real-source applicability. Use disclosed mock commercial stock and policies throughout.

### Authorized merchant pilot

Replace fictional listings with an explicitly authorized connector or merchant-maintained feed. Record seller identity, permissions, freshness and actual policy authority. Public storefront scraping alone does not establish private quantity, unit condition, cost, delivery capacity or discount authority. Show an indicative quote until the merchant can commit its stock and terms.

Support read-only connection first, then quote permission, then reservation/order permission through separately tested adapters. Do not treat a connected catalog as permission to change prices or fulfill orders. Retain the same offer, approval and execution contracts.

Pilot merchant-provided costs or floors; merchants need not expose full accounting to buyers or rivals. Measure repeat participation, contribution and incremental demand before training an alleged optimal-pricing model. The platform must not pool confidential future strategies across sellers to coordinate prices.

### Broader category and merchant coverage

Add profiles and real evidence sources with positive/negative applicability tests. Clothing needs explicit sizing/measurement conventions; appliances need electrical and installation scope; household products need interface/dimension certainty. Scale matching before expensive generation: category, service area, condition, quantity and deadline first, then relevant merchants.

At greater scale, use indexed/semantic retrieval where it measurably helps, durable fan-out, quota/cost allocation, immutable audit history and provider-specific connectors. External merchant agents can later use a narrow typed quote protocol with authenticated identity and enforced policy; the prototype's internal context ID is not a public agent-verification standard.

### Later extensions requiring separate product decisions

Cross-merchant basket optimization; group buying; real delivery/installation capacity; order tracking and customer support; returns/refund reconciliation; voice/photo intake; merchant approval queues; external agents; learned price-response models; subscription or transaction fees. Each adds distinct authority, data and support work. A single buyer approval screen does not create atomic multi-seller settlement or escrow.

**What will remain true:** prompt-first simplicity, manufactured-product breadth, merchant-controlled commercial limits, actual competition rather than staged rhetoric, buyer-private ceiling, transparent scoring, independent applicable evidence and approval of exact final terms.

**Recommended next action:** M0 contract freeze between A/B and C/D, followed by one complete simulated purchase. Prove the state/permission boundary before expanding sponsor claims. The absence of real merchant permission is resolved for the mock demo, not for a production launch.

# 26. External dependencies and questions to resolve

| Dependency | Status / safe behavior |
|---|---|
| Merchant authority | Resolved for prototype: fictional merchants/stock approved and labeled. No live seller permission needed for this demo. |
| Provider accounts | Convex/Gemini/Apify access and quotas untested. Actual use must be demonstrated; fixtures/contracts can be built while obtaining keys. |
| Source coverage | Exact Actor/source pairs need testing for laptop and genuine non-tech item. Real cache/unavailable is valid; synthetic reviews prohibited. New category profiles need field/provenance tests. |
| Visa | Operation/access unknown; simulation reliable default. Enable only a tested, accurately labeled capability. |
| Figma / eligibility | A/B own design/UI. Confirm Figma access, exact judging proof, Agent access if targeted, category/award limits and Best Actor publication rules. No prize eligibility guaranteed. |
| Deployment | Confirm sponsor accounts/credits and hosting access; no spend/public deployment performed here. |

### Questions with assigned timing

- M0, D: What Visa sandbox operation/access is available, and can it establish the exact quote amount, idempotency and terminal status? Use simulation unless verified.
- M0, D: Which source-specific Actors and lawful accessible sources provide real applicable product/owner evidence? Which exact non-tech product is supported? Preserve honest unavailable states.
- M0, A/B: Which Figma categories, Agent access and design/proof artifacts are accepted? Confirm sponsor category caps and award combinations with organizers.
- M0, C/D: Measure model availability, source latency, provider quotas and cost. The selected model and caps are configurable starting choices, not performance guarantees.
- Pilot, C: Will actual merchants authorize discounts and stock/order access, and is contribution still positive after real costs and fees?
- Pilot, all: Will enough buyers submit useful requests and enough merchants repeatedly respond? Can this beat fixed-price search plus ordinary promotions?

**Readiness verdict:** product flow, role boundaries, commercial mechanism, data/state contracts and proposed implementation defaults are specified enough to begin the prototype. Provider operations and actual runtime behavior still require tests. No document can certify zero future implementation gaps.

**Verification boundary:** source/document review and PDF layout checks only. No application code, model request, Actor extraction, deployed auth, reservation, pricing model calibration or Visa operation has been executed. Test cases describe required outcomes, not observed passing application tests.

**TLDR:** v4 consolidates the prompt-first multi-category marketplace, merchant incentives and bounded bidding, neutral fit and independent sentiment, Figma responsibilities, implementation safeguards and sponsor/prize strategy. Begin with shared contracts and one end-to-end simulated path; access, evidence coverage, eligibility and real market economics remain open.

# 27. Source ledger and evidence limits

- S1: Vite deployment, environment, Convex actions, Workflow, scheduler.
- S2: Convex Auth setup, Anonymous, Password, authorization. No client-controlled role assignment.
- S3: Gemini 3.6 Flash, function calling, structured outputs. Shape does not establish truth.
- S4: Apify async runs, running Actors, webhooks. Source access/cost/output need testing.
- S5: Visa Intelligent Commerce, quick start. Tokenization alone is not checkout.
- S6: Event requirements, Fetch AI hackpack. ASI:One requirement conflicts with chosen UI; award stacking unconfirmed.
- S7: Apple MacBook Air M2 2022 US specs.
- S8: Apple MacBook Air 13-inch M3 2024 US specs.
- S9: Apple MacBook Air 13-inch M4 2025 US specs.
- S10: Lenovo US 21ML0053US. Specifications, not availability.

### Research and market references

- R1: Gartner shopping-AI research. Two separate surveys; avoid generalizing subgroup results.
- R2: Baymard checkout research. Usability findings, not a conversion forecast for this project.
- R3: Selected durability discussion. Anecdotal, unverified accounts; no representative social-listening sample.
- R4: Selected clothing-size discussion. Anecdotal fit concerns.
- R5: FTC individualized pricing study. Initial staff findings, not a legal ruling about BuyerUnion.
- R6: Auction theory overview. Mechanism background, not a proof of this marketplace's equilibrium.
- S11: Spacetime event announcement. Special prize mentioned; exact category/amount unconfirmed.
- C1: Alibaba RFQ.
- C2: Amazon Business RFQ. Prior review; current refresh returned 403.
- C3: Carwow partner model.
- C4: Google Direct Offers.
- C5: Shopify Agentic Storefronts announcement.
- C6: Pactum tactical sourcing.

Sponsor prize details principally use the two supplied event snapshots; their heading/description conflicts are preserved. Official event/Fetch/Spacetime pages provide current cross-checks, not confirmation of eligibility or award stacking. Original research reports remain background: visa-shopping-research.md and buyerunion-marketplace-report.md. Older laptop-first recommendations in those reports are superseded here.